PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

NORTHERN ZONE

FORM FOUR MOCK EXAMINATION 2026

032/2B **CHEMISTRY 2B**

THREE HOURS ADVANCED INSTRUCTIONS

IMPORTANT

1. Great care must be taken not to divulge these to both candidates and unauthorized persons either directly or indirectly
2. Preparation and labelling of chemicals

2.1 Question 1

- Prepare 0.1 mol/dm^3 of hydrochloric acid, allow 150cm^3 per candidate and label it A
- Prepare 0.05 mol/dm^3 of sodium carbonate decahydrate ($Na_2Co_3.10H_2O$) allow 150cm^3 per candidate and label it B
- Provide each candidate with methyl orange (MO) indicator
- Provide each candidate with 1 burette
- 1 titration flask, 1 pipette (20cm^3/25Cm^3)
- 1 white tile, 1 dropper, 1 retort stand and clamp.

2.2 Question 2

Prepare the following

- A solution of 0.13M of $Na_2 S_2 O_3$
  (Sodium thiosulphate solution) and allow 150 Cm^3 per candidate and label it MM.
- A solution of 2M HCL, allow 150cm3 per candidate well label it NN
- Thermometer (1 – 100°C) 1 per each candidate
- Heat source (burner) for sharing 1:4
- Stop watch, 1 per candidate
- A piece of white paper labelled X
- Prepare water bath using 800cm^3 beaker for sharing 1:4 place it on the heat source.
- Provide 100 cm^3 beaker per candidate
- Provide 10cm^3 measuring cylinder per each candidate
- Stirring glass rod, 1 per each candidate